Mini-Rayyan Hackathon: Pre-Read for Teams

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The task

You will build a small version of what Rayyan.ai does, end to end, in 72 hours. A researcher searches PubMed and other free databases from inside your tool, using their APIs, with different keyphrases, keyphrase combinations and date ranges. The researcher picks the searches that look useful and pulls their abstracts into the tool. Your tool removes duplicate abstracts, excludes studies that do not belong (for example non-human, simulation or non-English studies), and extracts PICO (Population, Intervention, Comparator, Outcome) from every abstract that remains. A person stays in charge: the tool lets a human override duplicates, exclusions and PICO tags, and logs each change.

Who: teams of 3 to 4 with mixed years (UG1, UG2, UG3), ideally led by UG2 or UG3 students. **Time:** 72 hours. **Compute:** free Google Colab and free-tier hosting.

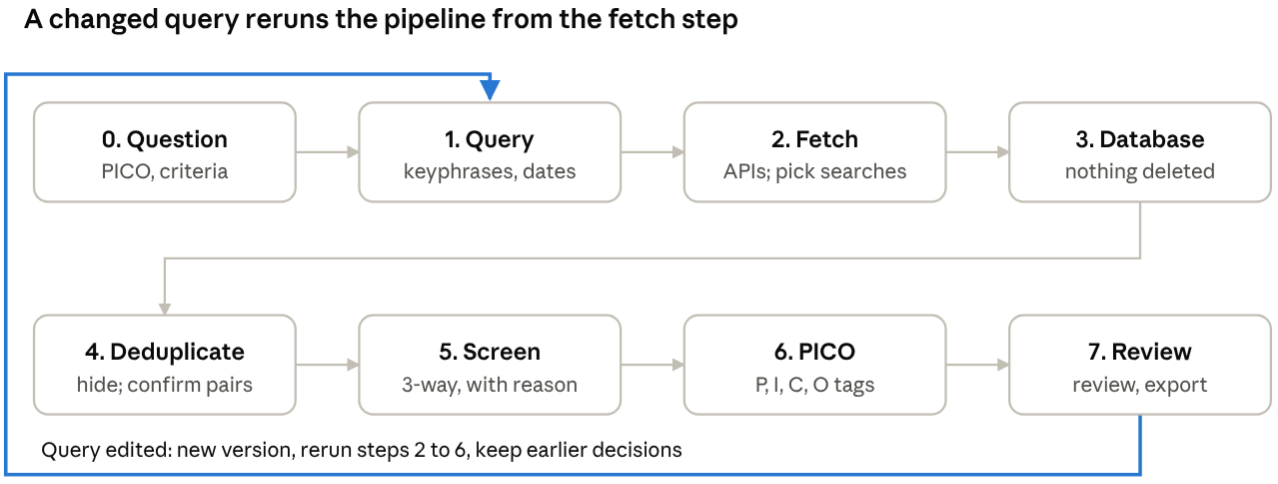
Why it is hard: each step is simple on its own. The challenge is the loop. When the query changes, the abstracts change, so deduplication and screening must be rerun without losing the work a reviewer has already done.

You must deliver: a working tool with a database, a method for each step and human-in-the-loop controls; a short report that documents your understanding of the problem; a one-page loop design (the Level 2 goal); and a five-minute demo. How you are scored is in the separate Scoring Metrics document.

Ground rules

- **Working beats clever.** A simple method that runs end to end beats a complex one that does not. Once it runs, improve each step as far as you can.
- **Any method is allowed.** Rules, classical machine learning or large language models (LLMs); AI is not required. Justify your choice.
- **Humans decide.** At a minimum, a person can override duplicates, exclusions and PICO tags, and the tool logs each change.
- **Understanding is graded.** Document the problem, the loops between steps and your design choices.
- **Your stack, your choice.** Programming language, database design and which literature databases to search are up to you. More sources are better, but a tool that works end to end comes first. LLMs may be used for code and documentation.

Pipeline



A human confirms or overrides at every decision step.

Mini-Rayyan pipeline · steps 0 to 7, 1 loop

The accent line is the Level 2 loop (the design is required, the implementation is a bonus): editing the query creates a new version and sends the system back through the fetch step, and the review step must keep every decision the reviewer already made. Step 0 is the review question and criteria; steps 1 to 7 are in the Level 1 table below.

Step 0: your review question

Before any search, write one page with the review question in PICO form and the inclusion and exclusion criteria, each with a one-line reason. Keyphrases come from the question, and screening applies these criteria. Bring this page to the hour 12 checkpoint.

Example (illustrative only; choose your own topic):

Part	Example
Population	Adults with type 2 diabetes
Intervention	Metformin
Comparator	Placebo or another glucose-lowering drug
Outcome	Change in HbA1c (glycated haemoglobin)

Review question: in adults with type 2 diabetes, does metformin, compared with placebo, lower HbA1c? Keyphrases come from it: "type 2 diabetes", "metformin", "placebo", "HbA1c".

- Include human studies of adults, because the question is about adults.

- Include trials that compare metformin with a comparator, because those can answer the question.
- Exclude animal studies, because the question is about people.
- Exclude simulation studies, because they model results and do not measure patients.
- Exclude non-English text, because the team cannot check it.
- Exclude case reports, because one patient cannot show an average effect.

These exclusion reasons are examples only. Teams propose and justify more from their own literature search.

Level 1: end-to-end system

Every team must finish Level 1: a working UI, a database and a method for each step, with a human able to override duplicates, exclusions and PICO tags. Design for Level 2 from the start so you do not rebuild later: store every output with the query version that produced it, and keep human decisions apart from machine results.

Nothing is deleted: duplicates and excluded records are hidden and marked, never removed from the database, so every decision can be traced and undone.

The examples below are invented to show the format; your records will differ.

Step	What to build	Basic method or rule	Example
1. Query	A form for keyphrases, keyphrase combinations (AND, OR, NOT) and a date range. Run several searches, one at a time; each is saved as a query version with its hit count so the researcher can choose which to pull.	None	("type 2 diabetes" OR "type II diabetes") AND metformin AND placebo, 2015 to 2025, saved as version 1. A second search, metformin AND sulfonylurea, is saved as version 2.

Step	What to build	Basic method or rule	Example
2. Fetch	Search from inside the tool through the APIs of PubMed, Europe PMC or other free databases, respecting their limits. After the researcher approves a search, pull its abstracts; store title, abstract, authors, year, journal, identifiers and source.	None	The researcher sees the hit counts for versions 1 and 2, approves both, and the tool pulls their abstracts, storing the source with each record.
3. Database	Tables for queries, records and reviewer decisions.	None	A table links each record to the query versions that found it, so a record found twice is stored once.
4. Deduplicate	Group records that are the same paper, within a search and across the outputs of all searches run so far, and keep one per group. Hide the other records; never delete them. A human confirms uncertain pairs. Further reports of one study (for example a journal paper and its conference version) are not duplicates.	Exact match on DOI or PMID when present. When the DOI is missing, build and document an algorithm that compares several fields (such as cleaned title, year, first author, journal and pages).	Versions 1 and 2 both return one trial. One record has a DOI and the other does not, but the title (ignoring capitals), year and first author match, so they form one group: one is shown, one is hidden.

Step	What to build	Basic method or rule	Example
5. Screen	Apply the criteria from step 0 and mark each record Include, Exclude or Maybe; every exclusion needs a reason. For each criterion, use the record's metadata field when it has one; otherwise build and document an algorithm to detect it. The reviewer chooses which criteria are active and can override any exclusion.	Rules and manual review are enough. A relevance classifier, such as TF-IDF with logistic regression, is optional.	A rat study: Exclude, reason "non-human". A German abstract: Exclude, reason "non-English". A pilot with an unclear population: Maybe.
6. PICO	For every included abstract, tag spans as Population, Intervention, Comparator or Outcome; the reviewer can correct tags.	Keyword and pattern rules are enough. A tagger trained on EBM-NLP is optional. Comparator needs your own rules.	"Adults with type 2 diabetes received metformin or placebo for 12 weeks; the primary outcome was change in HbA1c." Population: adults with type 2 diabetes. Intervention: metformin. Comparator: placebo. Outcome: change in HbA1c.
7. Review	Result list, Include, Exclude or Maybe with a reason, highlighted PICO spans, CSV export of included studies.	None	The reviewer changes a Maybe to Include, adds a note, and exports the included studies as a CSV.

Level 2: loops when the query changes

Required: before coding, hand in a one-page loop design, which is the Level 2 goal. For each step, say what a person might go back to change, which later steps that affects, and how reviewer decisions stay safe. Design Level 1 so it can grow into Level 2 without a rebuild.

Bonus: implementing the loop and passing the tests described in the Scoring Metrics document. Bonus points go to query versions that keep reviewer decisions, the diff view and the exact revert; incremental fetch, re-screening and the run log support them.

- **Query versions:** every edit creates a new version; old versions are kept and can be reopened.
- **Incremental fetch:** only fetch records not already stored, and remember which query versions found each record.
- **Rerun deduplication on the combined set:** a new record can join two groups that were separate before, or a group can change shape. Each group keeps one representative record.
- **Re-screen and re-tag only what is new or changed.**
- **Keep reviewer decisions:** a decision belongs to the group, not the query. When groups merge and the decisions conflict (one included, one excluded), show the conflict to the reviewer instead of choosing silently.
- **Diff view:** after each query edit, show records added, records removed and groups newly merged.
- **Repeatability:** reverting to the first query version must give the first record set and the same groups again.
- **Run log:** record which version, step and time each run used.

Questions your design must answer: what happens when a record you excluded appears again under a new query? What happens when two records you judged differently turn out to be duplicates?

Data and sources

Need	Free source or data	What to know
Search and abstracts	PubMed E-utilities	EFetch returns abstracts. Limit is 3 requests per second without an API key and 10 with one. Register a tool name and email with NCBI, show NCBI's disclaimer, and run large jobs on weekends or between 9 pm and 5 am US Eastern. Abstracts may contain copyrighted material.
Another free source	Europe PMC REST API (documented via the rOpenSci client)	No registration or API key. Searches return 100 records by default. Queries are expanded with MeSH synonyms unless switched off.
Relevance screening (optional)	SYNERGY	26 systematic reviews, 169,288 records, of which 2,834 (1.67%) were included. Released under CC0 1.0 according to the repository page; see also the DOI record.
PICO training and test (optional)	EBM-NLP (Hugging Face card)	4,993 abstracts annotated for Participants, Interventions and Outcomes, with no separate comparator label. Training labels come from crowd workers, test labels from medical professionals. The licence is listed as unknown, so for this event use it for training and testing only and do not redistribute it. The loading script must be converted to parquet.
Language and study-type filters	Fields inside the PubMed records (language, publication type)	If a field is missing, the team detects it with its own algorithm.

To be announced by the organizers at kickoff: how the frozen snapshot of results for testing without live servers is delivered and in what format, and how judges will reach your running tool. Judges also test with queries and records you do not see in advance. You use the live APIs for the demo.

What to hand in and when

1. **Hour 12:** data sources understood, review question and criteria written, one-page loop design drawn, database designed, steps 1 to 3 running on a small query.
2. **Hour 36:** Level 1 complete, with deduplication, screening and PICO baselines working end to end.
3. **Hour 60:** Level 2 loop attempted if time allows, report drafted.
4. **Hour 72:** submit a code repository with a README that lets a stranger rerun the work, a report of at most three pages, and a five-minute demo.

The report answers two questions. What was the simplest approach you tried first, and why did you keep or replace it? Was each added step worth the effort and compute, set against the reviewer time it saves? Label every figure as an estimate and list your assumptions. The report also documents your understanding of the problem: your design choices, which exclusion criteria you used, why, and whether each came from metadata or your own algorithm, and where a human decides.

To be announced by the organizers: where and how to submit, and whether the demo is live or recorded.

Abbreviations

- **API:** Application Programming Interface
- **CC0:** Creative Commons Zero (public domain dedication)
- **CSV:** Comma-Separated Values
- **DOI:** Digital Object Identifier
- **EBM-NLP:** Evidence-Based Medicine, Natural Language Processing (corpus)
- **EFetch:** Entrez Fetch (PubMed record retrieval utility)
- **HbA1c:** Glycated haemoglobin (a blood-sugar measure)
- **LLM:** Large Language Model
- **MeSH:** Medical Subject Headings
- **NCBI:** National Center for Biotechnology Information
- **PICO:** Population, Intervention, Comparator, Outcome
- **PMID:** PubMed Identifier

- **SYNERGY:** Open dataset on study selection in systematic reviews
- **TF-IDF:** Term Frequency, Inverse Document Frequency
- **UG1, UG2, UG3:** Undergraduate year 1, 2 and 3
- **UI:** User Interface